Fiscal Year:	FY 2008	**Task Last Updated:**	FY 06/25/2008
PI Name:	Schiestl, Robert H Ph.D.		
Project Title:	Effects of Space Radiation on Degenerative Tissue Disease, Genetic Instability and Oxidative DNA Damage in Ataxia Telangiectasis Deficient Mice		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	90095-8347	**Congressional District:**	33
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2004 Radiation Biology NNH04ZUU005N
Start Date:	08/15/2005	**End Date:**	08/14/2010
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 8/14/2010 per A. Chu/ARC (8/09) NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes then, per JSC/A. Chu-ARC (jvp 4/2009)		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Teitell, Michael (University of California, Los Angeles, Pathology Dept)		
Grant/Contract No.:	NNA05CV34G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Atm (ataxia telangiectasia mutated) heterozygosity has a very high prevalence in the human population and seems to be a risk factor for cardiac damage and cancer. Radiation also causes reactive oxygen species, which also seem to be involved in the etiology of Ataxia Telangiectasia, cardiac damage, movement disorder and cancer. Oxidative DNA damage, DNA strand breaks and genetic instability may be involved in the etiology of these diseases. Atm deficient homozygous patients and mice show an elevated frequency of DNA deletions and signs of oxidative stress in their cells. Preliminary results show that DNA deletions at the pun locus are induced by ionizing radiation. The DNA deletion assay is based on the quantification of black spots on fur and eyes resulting from reversion of the pun mutation. This reversion occurs by deletion of 70 kb of an internal duplication within the p gene. Aim 1 is to determine whether AtmDSRI homozygous or heterozygous mice may show an increased frequency of space radiation induced DNA deletions. We also found that the elevated frequency of DNA deletions in Atm truncation deficient mice is completely reverted to wildtype levels by nutritional supplementation with the radical scavenger and glutathione precursor N-acetyl cysteine (NAC). Thus, we will test in aim 2 whether AtmDSRI homozygous or heterozygous mice nutritionally supplemented with NAC may show reduced frequencies of space radiation induced DNA deletions whereas mice supplemented with the prooxidant and inhibitor of glutathione metabolism, BSO may show an increased frequency of radiation induced deletions, which will be tested in aim 2. In aim 3 we will test whether AtmDSRI homozygous and heterozygous mice may show an increased frequency of space radiation induced cardiac damage or cancer. Aim 4 is to test whether AtmDSRI homozygous and heterozygous mice may show an increased frequency of space radiation induced ataxia. Implications for human health may include that Atm missense mutation heterozygosity (a significant proportion of the human population) may cause an increased risk for space radiation induced cardiac damage ataxia or cancer and that potential astronauts may benefit from being tested for Atm heterozygosity and that chemoprevention with antioxidants may reduce the space radiation induced genomic instability and possibly cancer in Atm heterozygous people.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Implications for human health may include that Atm missense mutation heterozygosity (a significant proportion of the human population) may cause an increased risk for space radiation induced cardiac damage, ataxia or cancer and that potential astronauts may benefit from being tested for Atm heterozygosity and that chemoprevention with antioxidants may reduce the space radiation induced genomic instability and possibly cancer in Atm heterozygous people.
Task Progress:	ATM is a key gene involved in the cellular response to ionizing radiation induced DNA damage. The frequency of heterozygous ATM mutant carriers in the United States population may be as high as 1 in 100. Our research seeks to investigate whether mice heterozygous mutant for ATM exhibit increased sensitivity to high LET space radiation induced DNA deletions and cancer. During our third year we had one run at NSRL and continued to gather data on the long-term effects of high LET (1 Gy, 1 GeV 56Fe) radiation on Atm mutant, heterozygous, and wildtype mice. Specifically, we looked at lifespan, data from necropsy reports, cardiovascular damage, and neuromotor function. In addition we looked at short-term effects of high LET versus low LET radiation on DNA damage in vivo. Although we have completed all of our exposures, we are still collecting and analyzing most of our long-term data. To date, our data indicate that exposure to high LET radiation does not affect lifespan in Atm mutant mice, but may have an effect on Atm heterozygous and wildtype mice. Atm mutant mice have a decreased lifespan with and without radiation compared to Atm heterozygous and wildtype mice. In addition, Atm heterozygous mice have a decreased lifespan compared to wildtype mice with and without radiation. We also see that to Atm heterozygous females seem to be more sensitive to 1 Gy, 1 GeV 56Fe than to Atm heterozygous male mice, in terms of lifespan. Our completed pathology reports on deceased mice are limited, and therefore it is difficult to draw conclusions, however we found that spontaneously Atm mutant mice have an increased cancer frequency compared to Atm heterozygous and wildtype mice. High LET radiation does not seem to affect cancer rates so far. When comparing short-term effects of high and low LET (gamma-rays) radiation in Atm mutant, heterozygous, and wildtype mice on DNA damage we found differences among genotypes and between the different qualities of radiation. In general we found that Atm mutant mice showed more sensitivity towards low LET radiation than wildtype mice and that high LET radiation damaged DNA to similar extents in all mice.
Bibliography Type:	Description: (Last Updated: 08/25/2020)
Abstracts for Journals and Proceedings	Yamamoto ML, Reliene R, Hacke K, Hafter K, Westbrook A, Trouiller B, Schiestl RH. "Effects of space radiation on degenerative tissue disease, genetic instability and oxidative DNA damage in Ataxia telangiectasia deficient mice." NASA Human Research Program Investigators' Workshop, League City, Texas, February 4-6, 2008. Abstracts, NASA Human Research Program Investigators' Workshop, February 4-6, 2008, p. 71. , Feb-2008
Abstracts for Journals and Proceedings	Yamamoto ML, Hafer K, Reliene R, Trouiller B, Kelly O, Bral M, Schiestl RH. "The Effect of Space Radiation on Genetic Instability, Neuromotor Function, Carcinogenesis, and Longevity in Atm Mutant Mice." 18th Annual NASA Space Radiation Investigators' Workshop, Rohnert Park, CA, July 13 – 15, 2007. Abstracts, 18th Annual NASA Space Radiation Investigators' Workshop, July 13 – 15, 2007, p. 78. , Jul-2007
Abstracts for Journals and Proceedings	Schiestl RH, Trouiller B, Bral M, Westbrook A, Hafer K, Hacke K, Reliene R, Yamamoto ML. "Effects of space radiation on degenerative tissue disease, genetic instability and oxidative DNA damage in Ataxia telangiectasia deficient mice." 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, PA, June 30 – July 2, 2008. 19th Annual NASA Space Radiation Investigators' Workshop, June 30 – July 2, 2008, p. 114. , Jun-2008